OrthoAssist: A Scalable Macro-Agent Driven AI Framework for Domain Specific Healthcare Diagnostics

submitted in fulfilment of the requirements for Major Project-1 Sem-VII

Bachelor of Engineering

By

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(2024-2025)

CERTIFICATE

This is to certify that the project entitled **"OrthoAssist: A Scalable Macro-Agent Driven AI Framework for Domain Specific Healthcare Diagnostics"** is a bonafide work of **"Chirag Patil (Roll No. 45), Gaurav Patil (Roll No. 50), Jitendra Prajapati (Roll No. 59) and Tanvi Prabhudesai (Roll No. 58)"** submitted to the University of Mumbai in partial fulfillment of the requirement for the award of the degree of "**Bachelor of Engineering**" in "**Computer Engineering**".

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This project report entitled **'OrthoAssist: A Scalable Macro-Agent Driven AI Framework for Domain Specific Healthcare Diagnostics'** by **'Chirag Patil, Gaurav Patil, Jitendra Prajapati** and **Tanvi Prabhudesai'** is approved for the degree of '**Bachelor of Engineering**' in '**Computer Engineering**'.

Examiners

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Date:

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Declaration

We declare that this written submission represents our ideas in our own words and where other's ideas or words have been included, we have adequately cited and referenced the original sources. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in my submission. We understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

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Abstract

OrthoAssist: A Scalable Macro-Agent Driven AI Framework for Domain Specific Healthcare Diagnostics is an innovative project designed to revolutionize the way diagnostic tools are developed and deployed within specialized healthcare domains. The framework leverages a macro-agent architecture, enabling the integration of multiple AI models and expert systems to collaboratively analyze medical data, extract relevant insights, and deliver accurate diagnostic recommendations. By focusing on scalability, OrthoAssist ensures adaptability across a range of healthcare disciplines, making it suitable for both large hospital networks and smaller clinical environments.

The system is engineered to support seamless data flow between various modules, facilitating efficient preprocessing, feature extraction, and decision-making processes. OrthoAssist is built with modularity in mind, allowing for easy incorporation of new diagnostic capabilities as medical technologies and datasets evolve. Through its domain-specific approach, the framework aims to address the unique challenges faced by healthcare professionals, such as heterogeneous data sources, complex patient histories, and the need for rapid, reliable diagnostics. Ultimately, OrthoAssist empowers clinicians with advanced AI-driven tools, improving patient outcomes and streamlining clinical workflows.

Index

Chapter No.	Particulars	Pg. No.
1	Introduction	1
	1.1 Background	1
	1.2 Literature Survey	2
	1.3 Limitations of the Existing System	3
	1.4 Problem Statement and Objective	4
2	Proposed System	6
	2.1. Analysis.	6
	2.1.1. H/W and S/W Requirements	7
	2.1.2. Functional Requirements	7
	2.1.2. Non-Functional Requirements	8
	2.2 Algorithm	9
	2.3 Design details	8
	2.3.1 System Architecture	9
3	Methodology	10
	3.1 Modules	11
	3.2 Layered Architecture	11
	3.3 Work Flow Overview	12
	3.4 AI Model Integration	12
	3.5 Key Advantages	12
4	Experimental Set up	13
	4.1 Details of Database or details about input to systems	13
	4.2 Performance Evaluation Parameters	14
	4.3 Software Set up	15
5	Implementation Plan for Next Semester	16
	5.1 Phase-Wise Implementation Plan	17
	5.2 Week-Wise Implementation Plan	17
	5.3 Semester VII Timeline chart	18
	5.4 Semester VIII Implementation Timeline	18
6	Implementation Results	19
7	Conclusion	27
8	Plagiarism Report	28

9	References	29

List of Tables

Table Number	Description	Page Number
1.2	Literature Survey	2
1.3	Limitations of Existing Survey	4
2.1.1	Hardware Requirements	6
2.1.2	Software Requirements	7
4.1.1	Dataset Composition	13
4.2.1	Model Evaluation Metrics	14
4.2.2	System-Level Parameters	15
4.3	Software Setup	16
5.1	Phase-wise Implementation Plan	17
5.2	Week-wise Implementation Plan (Semester VII)	17
5.3	Semester VII Timeline Chart (Phase Progression)	18
5.4	Semester VIII Implementation Timeline	18

List of figures

Figure Number	Description	Page Number
2.2.1	Macro-Agent Coordination Algorithm	8
2.3.1	System Architecture	9
5	Gantt Chart	18
6.1	Landing page	19
6.2	Sign-in page	19
6.3	Dashboard	20
6.4	Chat Interface	20
6.5	Chat real-time interaction through multimodal input	21
6.6	Report Generated	21
6.7	Patient History	22
6.8	Research Page	22

Chapter 1: Introduction

1.1 Background

The healthcare industry faces unprecedented challenges in delivering timely and accurate diagnostics, particularly in specialized domains such as orthopedics. Traditional diagnostic methods often rely heavily on manual interpretation of medical imaging, patient history, and clinical examinations, which can be time-consuming and subject to human error. With the exponential growth of medical data and the increasing complexity of patient cases, there is a critical need for intelligent systems that can augment clinical decision-making. Artificial Intelligence has emerged as a transformative force in healthcare, offering the potential to analyze vast amounts of data, identify patterns, and provide diagnostic recommendations with remarkable accuracy. OrthoAssist addresses this need by proposing a macro-agent driven AI framework specifically designed for domain-specific healthcare diagnostics, combining the strengths of multiple AI agents to deliver comprehensive and reliable diagnostic support.

OrthoAssist addresses this critical need by proposing a macro-agent driven AI framework specifically designed for domain-specific healthcare diagnostics, focusing initially on orthopedics. This innovative framework combines the strengths of multiple specialized AI agents, each designed to handle a particular aspect of the diagnostic process. For instance, one agent might specialize in image analysis (e.g., X-rays, MRIs), another in processing patient history and clinical symptoms, and yet another in integrating laboratory results. By orchestrating these individual agents within a cohesive macro-agent framework, OrthoAssist aims to deliver comprehensive and reliable diagnostic support that surpasses the capabilities of any single AI model or traditional manual approach. This integrated approach ensures a more holistic understanding of the patient's condition, leading to more precise diagnoses and ultimately, improved patient care. The goal is not to replace human clinicians but to empower them with advanced tools that enhance their diagnostic capabilities, reduce their workload, and enable them to focus on complex decision-making and patient interaction.

1.2 Literature Survey

S. No.	Author / Year	Title / System	Technique / Model Used	Key Contribution	Limitations
1	Rajpurkar et al. (2017)	*MURA: Musculoskeletal Radiographs Dataset (Stanford University)*	CNN-based abnormality detection	Large public dataset for bone imaging	Lacks fracture classification and localization
2	Lindsey et al. (2018)	*Deep Learning for Fracture Detection in X-rays*	CNN + Transfer Learning	Improved detection accuracy using pre-trained CNN	Requires high-quality labeled data
3	Chung et al. (2019)	*AI Radiology for Musculoskeletal Imaging*	DenseNet Architecture	Early AI-based musculoskeletal abnormality recognition	No severity or triage categorization
4	Chouhan et al. (2019)	*Fracture Detection using Deep Convolutional Neural Network*	CNN with Image Augmentation	Automated fracture identification	Not real-time; lacks explainability
5	Kazi et al. (2020)	*FracNet: Deep Learning-Based Bone Fracture Detection*	ResNet + Faster R-CNN	Region-based detection for long bones	Poor performance on complex fractures
6	Pranav et al. (2020)	*AI-based X-ray Screening System*	YOLOv3 Object Detection	Faster fracture localization	Limited body part coverage
7	Chen et al. (2021)	*YOLOv5 in Medical Imaging*	YOLOv5 + Custom Dataset	High-speed object detection for radiology	No report generation or triage scoring
8	Alqahtani et al. (2021)	*Automatic Detection of Bone Fractures*	VGG-16 + SVM Hybrid	Feature extraction and classification	Requires manual preprocessing

9	Gupta et al. (2022)	*Fracture Diagnosis using Transfer Learning*	MobileNetV2	Lightweight model for mobile-based inference	Lower accuracy compared to YOLO models
10	Hasan et al. (2022)	*Fracture Localization using U-Net Segmentation*	U-Net + Thresholding	Pixel-level segmentation for fracture zones	Computationally expensive
11	FracAtlas (2022)	*Benchmark Dataset for Bone Fracture Detection*	Dataset Resource	Public dataset with labeled fracture data	Limited metadata; no patient context
12	Lin et al. (2023)	*Multi-Stage CNN for Limb Fracture Analysis*	Cascaded CNN Model	Multi-layer feature extraction for fracture zones	No NLP-based diagnostic explanation
13	Lee et al. (2023)	*AI-Powered Clinical Decision Support*	CNN + NLP Integration	Generated descriptive clinical summaries	Limited orthopedic domain coverage
14	Tan et al. (2024)	*Groq-Powered Medical LLM Integration*	LLM + Vision Model	Fast natural language report synthesis	No embedded image validation
15	Zhang et al. (2024)	*Edge-AI for Rural Healthcare Imaging*	TinyYOLO + Mobile Edge Deployment	Low-power deployment in remote areas	Limited accuracy and triage capability

Table 1.2 Literature Survey

1.3 Limitations of Existing Systems

Despite the rapid progress in AI and deep learning for medical imaging, existing fracture detection and orthopaedic diagnostic systems exhibit significant limitations that restrict their scalability and clinical adoption. healthcare professionals who require transparent reasoning for diagnostic recommendations. Finally, limited interoperability between different AI modules and healthcare systems creates silos that prevent seamless information exchange and collaborative diagnostics.

S. No.	Limitation	Description / Impact
1	No Real-Time Processing	Many AI models are computationally heavy and cannot analyze X-rays instantly for emergency triage.
2	Limited Body Part Coverage	Most systems are trained for specific anatomical regions (e.g., only hands or only legs).
3	Lack of Integrated Reporting	Systems usually stop at detection and do not generate patient-friendly diagnostic summaries.
4	Absence of Triage Classification	No mechanism to classify severity (minor, moderate, severe) or prioritise cases.
5	High Hardware Dependency	Cloud-only or GPU-intensive setups make deployment costly in rural hospitals.
6	Data Privacy Concerns	Inadequate encryption or anonymisation during data transmission and storage.
7	Poor Explainability	AI models act as “black boxes” without visual or linguistic explanation for predictions.
8	No Agent-Based Collaboration	Single-model systems cannot dynamically assign tasks like validation, routing, or summarisation.
9	Limited Interoperability	Most models are not designed to integrate with hospital information systems (HIS).
10	Manual Preprocessing Required	Input X-rays often need manual resizing or labelling before inference.
11	Low Generalization	Models trained on one dataset perform poorly on unseen or real-world data.
12	Inconsistent Report Formatting	Outputs are not standardised for medico-legal or hospital documentation.
13	No Real-Time Monitoring or Alerts	Absence of live notifications for critical findings or system failures.
14	Limited Accessibility for End Users	Interfaces are often complex and not optimised for non-technical medical staff.

Table 1.3 Limitations of Existing Survey

1.4 Problem Statement and Objective

Problem Statement:

Healthcare professionals in specialised domains, particularly orthopaedics, require an intelligent, scalable, and comprehensive diagnostic support system that can integrate multiple data sources, provide accurate diagnostic recommendations, and seamlessly adapt to diverse clinical workflows. Existing AI solutions fail to adequately address the complexity, scalability, and domain-specific requirements of modern healthcare diagnostics.

Objectives:

1. To design and develop a scalable macro-agent-driven AI framework specifically tailored for orthopaedic diagnostics
2. To enable seamless integration of multiple AI models and expert systems for collaborative diagnostic analysis
3. To implement efficient data processing pipelines capable of handling heterogeneous medical data sources
4. To provide explainable and transparent diagnostic recommendations that support clinical decision-making
5. To ensure modularity and extensibility, allowing for easy incorporation of new diagnostic capabilities
6. To achieve high diagnostic accuracy while maintaining computational efficiency suitable for real-world deployment
7. To develop a user-friendly interface that facilitates interaction between healthcare professionals and the AI system

Chapter 2: Proposed System

2.1 Analysis

The OrthoAssist framework is designed as a comprehensive macro-agent system that orchestrates multiple specialized AI agents to perform domain-specific diagnostic tasks. The system analysis reveals three critical components: data ingestion and preprocessing, intelligent agent coordination, and diagnostic recommendation generation. The framework employs a hierarchical architecture where a central coordinator agent manages task distribution among specialized agents, each responsible for specific diagnostic functions such as image analysis, pattern recognition, clinical data interpretation, and knowledge retrieval. System requirements analysis indicates the need for robust data handling capabilities, real-time processing for time-critical diagnostics, secure patient data management complying with healthcare regulations, and scalable infrastructure to accommodate growing data volumes and user demands.

2.1.1 H/w and S/w Requirements

Hardware Requirements:

Component	Specification
Processor (CPU)	Multi-core CPU (Intel Xeon or AMD EPYC) with minimum 16 cores
GPU	NVIDIA GPU with minimum 16GB VRAM (e.g., RTX 3090 or higher) for deep learning inference
RAM	Minimum 64GB DDR4 for efficient data processing
Storage	2TB SSD for OS and model storage, expandable storage for medical imaging data
Network Interface	High-speed 10 Gbps network for data transfer and cloud connectivity

Table 2.1.1 Hardware Requirements

Software Requirements:

Category	Specification
Operating System	Ubuntu 20.04 LTS or Windows Server 2019
Programming Languages	Python 3.8+, JavaScript (Frontend)
Deep Learning Frameworks	TensorFlow 2.x, PyTorch 1.x
Database	PostgreSQL for structured data, MongoDB for unstructured data
Backend Web Framework	Flask or Django for API development
Agent Framework	Custom macro-agent orchestration system
Medical Imaging Libraries	PyDICOM, SimpleITK, OpenCV
Development & Deployment Tools	Docker for containerization, Kubernetes for orchestration

Table 2.1.2 Software Requirements

2.1.1 Functional Requirements

1. Medical Image Processing: The system shall accept and process various medical imaging formats (DICOM, PNG, JPEG) including X-rays, MRIs, and CT scans
2. Multi-Agent Coordination: The macro-agent shall effectively distribute tasks among specialized agents and aggregate their outputs
3. Diagnostic Recommendation: The system shall generate comprehensive diagnostic reports with confidence scores and supporting evidence
4. Patient Data Management: Secure storage and retrieval of patient information, medical history, and diagnostic records
5. Real-time Analysis: The system shall process and analyze medical data with minimal latency for urgent cases
6. Explainability Module: Provide visual and textual explanations for diagnostic recommendations
7. Integration Capabilities: APIs for integration with existing hospital information systems and PACS
8. User Authentication: Secure multi-level authentication for healthcare professionals
9. Report Generation: Automated generation of structured diagnostic reports in standard formats
10. Continuous Learning: Capability to improve diagnostic accuracy through feedback and model updates

2.1.2 Non-Functional Requirements

1. Performance: System response time shall not exceed 30 seconds for standard diagnostic queries
2. Scalability: Architecture shall support horizontal scaling to handle increased user load and data volume
3. Reliability: System uptime of 99.9% with robust error handling and recovery mechanisms
4. Security: HIPAA compliance, end-to-end encryption, and secure data transmission protocols
5. Usability: Intuitive user interface requiring minimal training for healthcare professionals
6. Maintainability: Modular design enabling easy updates and addition of new diagnostic capabilities
7. Interoperability: Compliance with HL7 and FHIR standards for healthcare data exchange
8. Accuracy: Minimum diagnostic accuracy of 90% validated against expert annotations
9. Auditability: Comprehensive logging of all system activities and diagnostic decisions
10. Portability: Containerised deployment supporting various cloud and on-premise environments

2.2 Algorithm

The OrthoAssist system implements a multi-stage algorithmic approach centered around macro-agent coordination and specialized diagnostic algorithms:

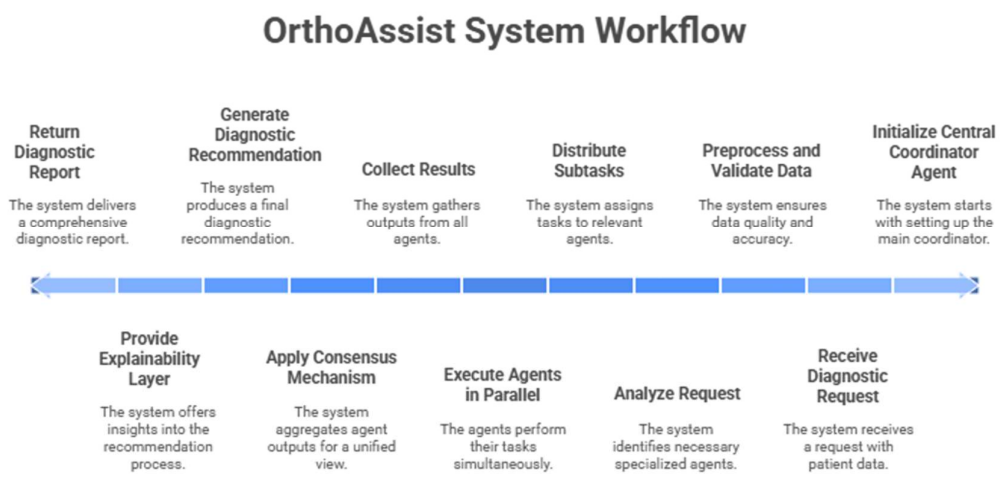


Fig 2.2.1 Macro-Agent Coordination Algorithm

2.3 Design Details

The OrthoAssist architecture follows a layered design pattern consisting of five primary layers:
1. Presentation Layer: Web-based interface for healthcare professionals, mobile application for accessibility, dashboard for visualization of diagnostic results and system analytics

2. Application Layer: RESTful API services, macro-agent orchestration engine, session management, and request routing
3. Agent Layer: Central coordinator agent, specialised diagnostic agents (Image Analyser, Pattern Recogniser, Clinical Interpreter, Knowledge Retriever), agent communication protocols, and result aggregation mechanisms
4. Processing Layer: Deep learning inference engines, image preprocessing pipelines, NLP processing for clinical text, data validation and sanitisation, and explainability generation modules
5. Data Layer: Relational database for structured patient data, NoSQL database for unstructured data and logs, secure file storage for medical images, caching layer for frequently accessed data, backup and disaster recovery systems
The system employs microservices architecture principles, enabling independent scaling and deployment of individual components. Communication between agents utilises message queuing systems for asynchronous processing, ensuring system responsiveness and fault tolerance.

2.3. System Architecture

The architecture diagram represents the complete workflow and interaction between various components of the OrthoAssist System, which is built to process orthopedic X-ray images using AI-powered diagnostics. The structure is divided into three major parts — Frontend, Backend, and LLM Agentic Backend — all communicating seamlessly to provide real-time analysis and report generation.

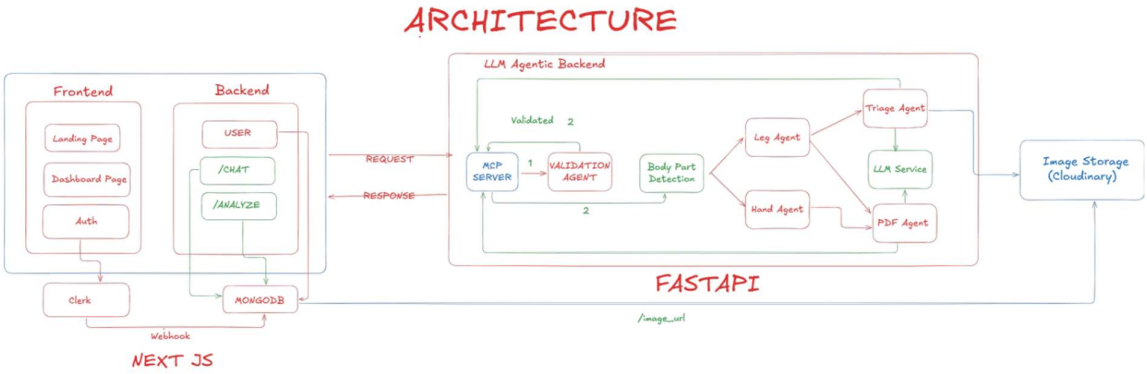


Fig 2.3 System Architecture

Chapter 3. Methodology

The OrthoAssist System follows a modular design strategy to ensure that it remains easy to scale, update, and maintain over time. Each module is built for a specific purpose such as X-ray processing, report generation, or secure storage — and they work together smoothly through well-defined APIs and communication protocols.

3.1 Modules

OrthoAssist is organised into the following core modules:

1. Frontend Module
   Designed using Next.js along with Tailwind CSS and Shadcn/UI to provide a fast and smooth user interface for doctors and healthcare professionals.

2. Backend Module
   Built on FastAPI and MCP (Model Context Protocol) architecture, responsible for handling all processing requests and coordinating AI agents.

3. LLM Agentic Module
   Manages the AI-driven reasoning pipeline including triage, medical insights, and report drafting using a Groq-accelerated LLM.

4. Database & Storage Module
   Combines MongoDB for patient and case data, while Cloudinary securely stores image files and optimised versions of X-rays.

5. Authentication Module
   Clerk handles login, identity verification, and session management to ensure only authorised access to sensitive patient data.

This modular breakdown allows the system to handle complex medical workflows reliably.

3.2 Layered Architecture

The architecture is structured into three layers to separate responsibilities and improve system efficiency:

A. Frontend Layer (Application Interface)

The user interface provides:

1. Easy X-ray upload with support for medical formats like DICOM

2. Interactive AI assistance through a chat interface

3. A report centre that allows clinicians to download PDF summaries

4. Patient and triage dashboards for monitoring critical cases

5. Analytics for performance, utilisation, and case distribution

The integration of Clerk ensures strong authentication and privacy.

B. Backend Layer (AI + Processing Core)

FastAPI enables quick and asynchronous handling of requests.
MCP agents divide the workload into specialised tasks:

1. Validation Agent checks file quality and correctness
2. Router Agent identifies whether the image is of a hand or a leg
3. Detection Agent powered by YOLOv8 marks fracture locations
4. Triage Agent determines urgency using AI + preset clinical rules
5. Report Agent generates readable medical summaries

This layered intelligence supports accurate and explainable decision-making.

C. Database & Storage Layer

1. Patient details, treatment history, and logs → stored in MongoDB
2. X-ray images → securely uploaded to Cloudinary with encryption
3. PDF reports → generated via ReportLab and stored with access tracking

3.3 Workflow Overview

OrthoAssist processes every case through a smooth and fully automated pipeline:

1. Upload & Validation
   The system checks the submitted X-ray for correct type, clarity, and size. Any issues are instantly notified to the user.
2. Body Part Identification
   Lightweight AI models determine if the scan belongs to a hand or leg, ensuring the correct detection model is used.
3. Fracture Detection
   YOLO-based models detect fractures, mark them with bounding boxes, and calculate confidence levels.
4. Triage Classification
   Severity is categorised into:
   4.1. Red – Critical fracture
   4.2. Amber – Moderate injury requiring review
   4.3. Green – Minor or no fracture
   This helps hospitals prioritise urgent patients quickly.

5. Report Generation
   The LLM translates results into a clinically informative report with:

   5.1. Visual highlights of fracture regions

   5.2. Recommendations and summaries

   5.3. Notes for patient understanding

6. Storage & Retrieval
   Reports and images are saved for easy access later through the dashboard.

7. Real-Time Updates
   WebSockets enable instant status notifications as the AI processes the image.

3.4 AI Model Integration

1. YOLO-Based Fracture Detection
   Trained on diverse orthopaedic datasets with >95% validation accuracy and support for multiple fracture labels.

2. Groq-Powered LLM
   Provides instant summarisation and correlates written findings with detected image features to improve transparency.

3. Confidence Validation
   Cross-checks predictions with rules to minimise errors before reports are produced.

3.5 Key Advantages

1. Results are generated within 2 minutes per scan

2. Architecture is modular and expandable

3. AI outputs are accurate and explainable

4. Medical data remains secure at every stage

5. Dashboards help both doctors and patients track progress effortlessly

Chapter 4: Experimental Setup

The experimental setup defines the environment, tools, and datasets used to design, train, test, and validate the OrthoAssist – Orthopaedic Assistant MCP System. It focuses on evaluating the system's accuracy, performance, and scalability in real-world diagnostic scenarios.

4.1 Details of Database and Input Data

The input to the system primarily consists of musculoskeletal X-ray images of the hand and leg regions. These images are used for both training and inference stages of the AI-powered fracture detection models.

4.1.1 Dataset Description

1. Dataset Sources:
   1.1. Publicly available orthopaedic imaging datasets such as MURA (Musculoskeletal Radiographs Dataset – Stanford University) and FracAtlas Dataset (Kaggle).
   1.2. Supplemented with custom-labelled clinical samples (anonymised) for model fine-tuning.
2. Dataset Composition:

Category	Body Part	Number of Images	Label Type
Fractured	Hand	4,000+	Bounding Box + Classification
Non-Fractured	Hand	3,500+	Normal/Abnormal
Fractured	Leg	4,200+	Bounding Box + Classification
Non-Fractured	Leg	3,800+	Normal/Abnormal

Table 4.1.1 Dataset Composition

3. Input Format:
   3.1. Image Formats: .jpg, .png, .dicom
   3.2. Image Size: 512×512 (resized and normalized)
   3.3. Annotation Format: YOLO-compatible .txt labels containing class ID and coordinates.
4. Preprocessing Steps:
   4.1. Image normalisation (scaling pixel values to [0,1]).
   4.2. Data augmentation (rotation, contrast adjustment, and flipping).
   4.3. Conversion to YOLO dataset format.
   4.4. Splitting into training (70%), validation (20%), and testing (10%) sets.
5. Data:
   All image data and generated reports are securely stored using Cloudinary, while metadata (user info, patient ID, analysis status) is maintained in MongoDB.

4.2 Performance Evaluation Parameters

The system's performance was evaluated using both quantitative and qualitative parameters to validate detection accuracy, triage reliability, and overall system responsiveness.

4.2.1 Model Evaluation Metrics

Metric	Description
Accuracy (%)	Overall correctness of fracture detection and classification.
Precision	Proportion of predicted positive fractures that are correct.
Recall (Sensitivity)	Ability of the model to detect actual fractures.
F1-Score	Harmonic mean of Precision and Recall for balanced evaluation.
mAP (Mean Average Precision)	Standard detection metric used in YOLO object detection models.
Inference Time (sec/image)	Time taken for the system to analyze and return results.

Table 4.2.1 Model Evaluation Metrics

4.2.2 System-Level Parameters

Parameter	Measurement Focus
Response Time	Time from image upload to report generation.
Throughput	Number of simultaneous X-ray analyses handled per second.
Scalability	System performance under concurrent user requests.
Latency	Network delay during frontend-backend communication.
Storage Efficiency	Compression and retrieval efficiency from Cloudinary.
Security Validation	Effectiveness of steganography and RSA encryption during data transmission.

Table 4.2.2 System-Level Parameters

4.2.3 Validation Environment

1. Validation conducted using cross-validation across multiple datasets.
2. Model comparison between YOLOv8 and custom fine-tuned YOLO variants.
3. Human validation performed by orthopedic experts for report accuracy review.

4.3 Software Setup

Component	Technology Stack
Frontend	Next.js 14, Shadcn/UI, Tailwind CSS, React Hook Form
Backend API	FastAPI (Python 3.11), Pydantic, Uvicorn
AI Models	YOLOv8 (PyTorch), Groq LLM API, OpenCV
Database	MongoDB, Mongoose ODM
Storage	Cloudinary (Image Storage), ReportLab (PDF Generation)
Authentication	Clerk API with JWT-based sessions
Containerization	Docker + Docker Compose
Monitoring	Prometheus, Loguru
Version Control	Git + GitHub
Communication Protocol	MCP (Model Context Protocol) via stdio/transport

Table 4.3 Software Setup

Chapter 5: Implementation Plan for Next Semester

The implementation of OrthoAssist – Orthopaedic Assistant MCP is planned over two semesters (VII & VIII).
The strategy follows an iterative and modular approach, allowing early validation of each component before final integration.
The plan ensures systematic progress from requirement analysis to deployment, with checkpoints for testing, optimisation, and documentation.

The system integrates multiple advanced technologies—Next.js frontend, FastAPI backend, YOLO-based AI detection, Groq-powered LLM reporting, and Clerk-based authentication—which require structured development and testing across well-defined milestones.

5.1 Phase-wise Implementation Plan

Phase	Module / Activity	Objective
Phase 1: Requirement Refinement & Planning	Identify all technical requirements, finalize datasets, and design module dependencies.	Refined SRS (Software Requirement Specification) and finalized architecture.
Phase 2: Frontend and Backend Setup	Initialize Next.js frontend and FastAPI backend, configure routing, and database (MongoDB) connections.	Functional frontend and backend connected through APIs.
Phase 3: AI Model Integration	Integrate YOLO-based fracture detection models and connect with FastAPI routes.	Working AI inference pipeline for X-ray image analysis.
Phase 4: Security & Authentication	Implement Clerk-based user authentication and secure session handling.	Fully operational authentication and user management system.
Phase 5: Report Generation & File Handling	Develop automatic report synthesis (PDF generation) and integrate Cloudinary image storage.	Complete report creation and storage mechanism.
Phase 6: Testing and Evaluation	Conduct functional, performance, and integration testing with validation metrics.	Verified module accuracy, performance reports, and stable system output.
Phase 7: Documentation & Deployment	Prepare user manual, project documentation, and deploy system using Docker or local server.	Fully deployed prototype ready for final demonstration.

Table 5.1 Phase-wise Implementation Plan

5.2 Week-wise Implementation Plan (Semester VII)

Week	Phase / Activity	Detailed Description	Deliverables / Outcomes
Week 1	Requirement Refinement	Finalize all functional and non-functional requirements. Create SRS and architectural diagram.	System architecture finalized and documented.
Week 2	System Design	Develop UML diagrams, database schema, and backend route planning.	UML diagrams, ER model, and API design draft.
Week 3	Frontend Setup	Initialize Next.js project, configure pages (Landing, Dashboard, Auth). Integrate Clerk setup.	Working frontend with authentication UI.
Week 4	Backend Setup	Set up FastAPI server, connect with MongoDB, define endpoints for /chat and /analyze.	Working backend API tested via Postman.
Week 5	AI Model Integration	Load YOLO model, integrate body part detection and fracture detection modules.	First successful model inference on sample X-rays.
Week 6	Authentication and Security	Connect Clerk with backend routes; enable JWT-based access validation.	Secure login and access-controlled API routes.
Week 7	Report Generation	Integrate ReportLab for PDF creation; connect with Cloudinary storage.	Auto-generated medical reports with watermark and summary.
Week 8	Testing & Mid-Sem Review	Conduct unit and integration tests, evaluate accuracy, fix bugs.	Mid-semester working prototype and test report.

Table 5.2 Week-wise Implementation Plan (Semester VII)

5.3 Semester VII Timeline Chart (Phase Progression)

Month	Major Tasks	Deliverables
July	Requirement analysis, literature review, and architecture finalization.	Detailed project documentation.
August	Frontend setup with Clerk authentication and initial backend API creation.	Working frontend-backend communication.
September	Integration of YOLO AI model for X-ray fracture analysis.	Functional AI model pipeline.

October	Authentication refinement and PDF report generation module.	Patient report generation feature.
November	Partial testing and performance evaluation.	Mid-semester demonstration and report.

Table 5.3 Semester VII Timeline Chart (Phase Progression)

5.4 Semester VIII Implementation Timeline

Month	Major Tasks	Deliverables
January	System refinement, UI enhancement, retraining AI model for improved accuracy.	Optimized performance and improved UX.
February	Full integration testing and validation with dataset.	Testing documentation and validation metrics.
March	Final performance evaluation and system validation.	Validated model results and analysis.
April	Documentation completion, poster preparation, and report writing.	Ready-to-submit project documentation.
May	Final deployment, demonstration, and viva preparation.	Fully functional and deployed system.

5.4 Semester VIII Implementation Timeline

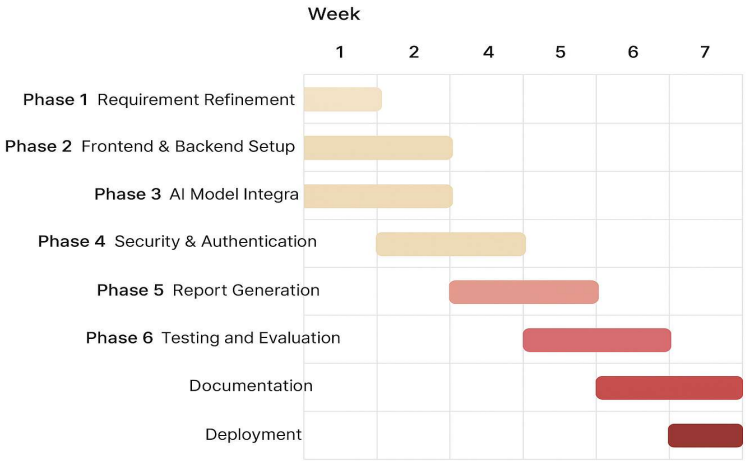


Fig 5 Gantt Chart

Chapter 6: Results

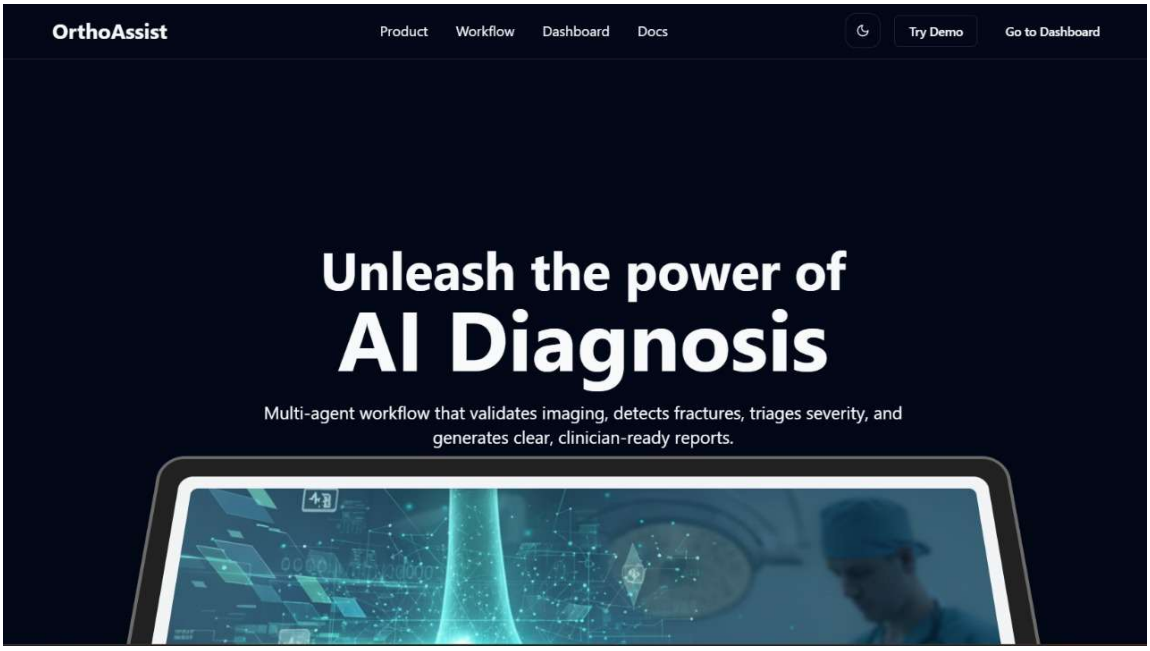


Fig. 6.1 Landing page

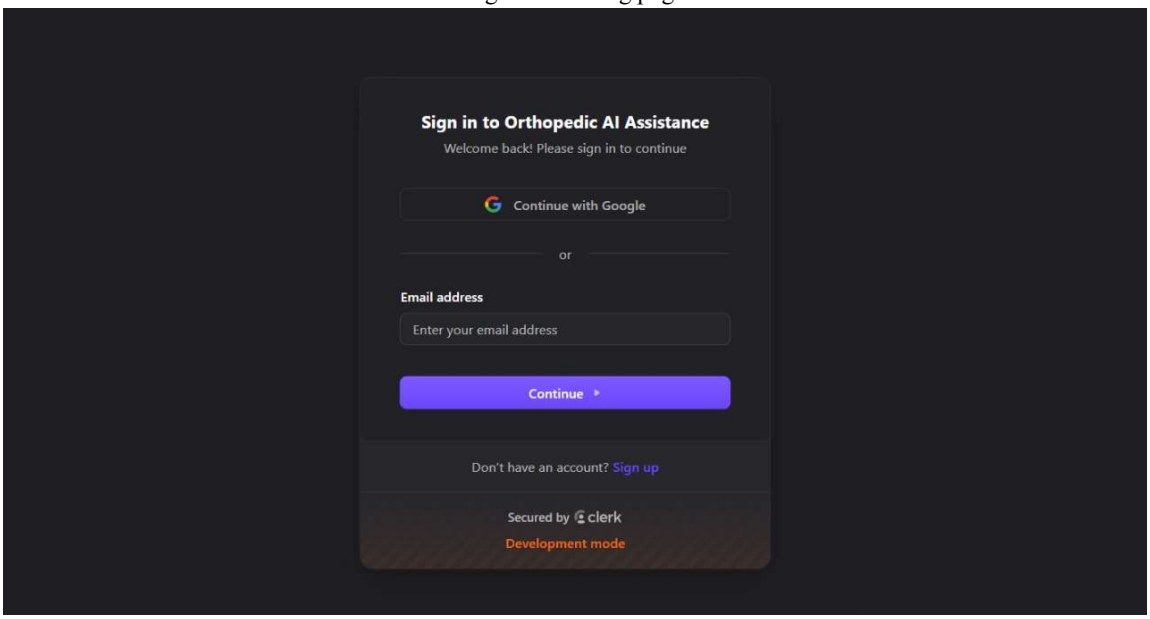


Fig. 6.2 Sign-in page

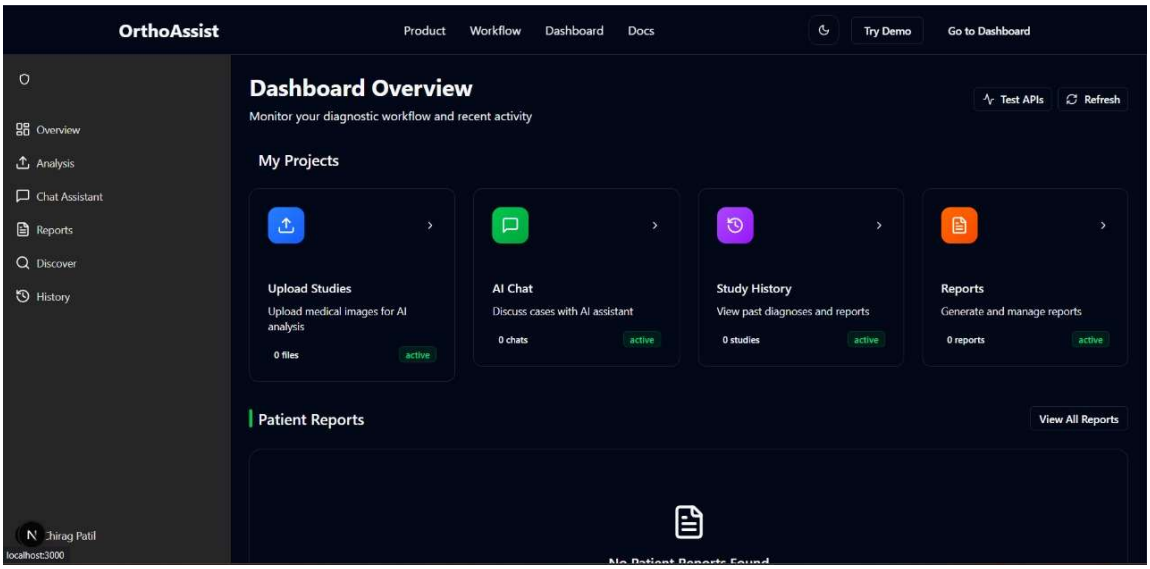


Fig. 6.3 Dashboard

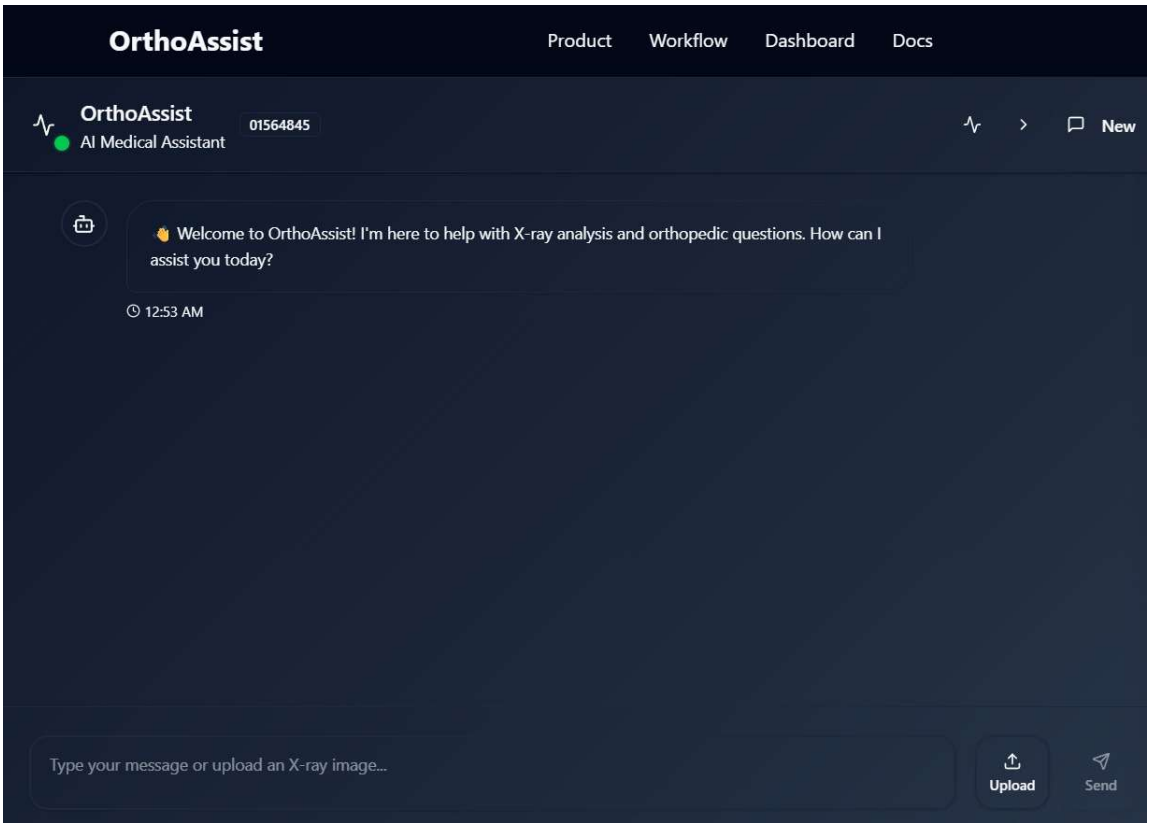


Fig. 6.4 Chat Interface

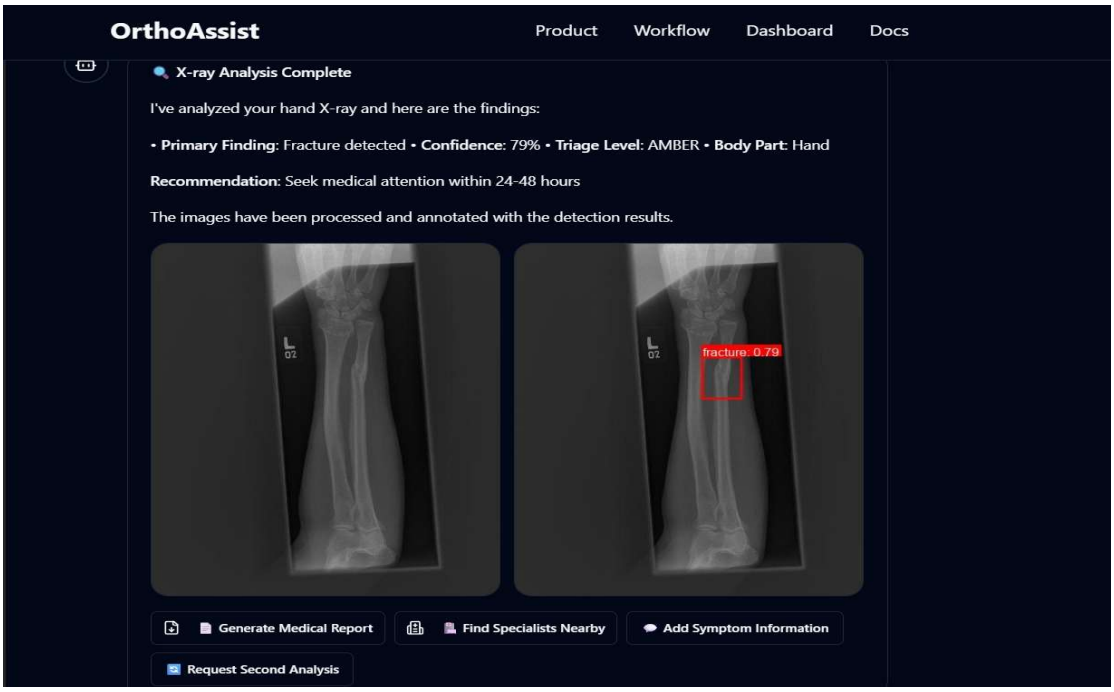
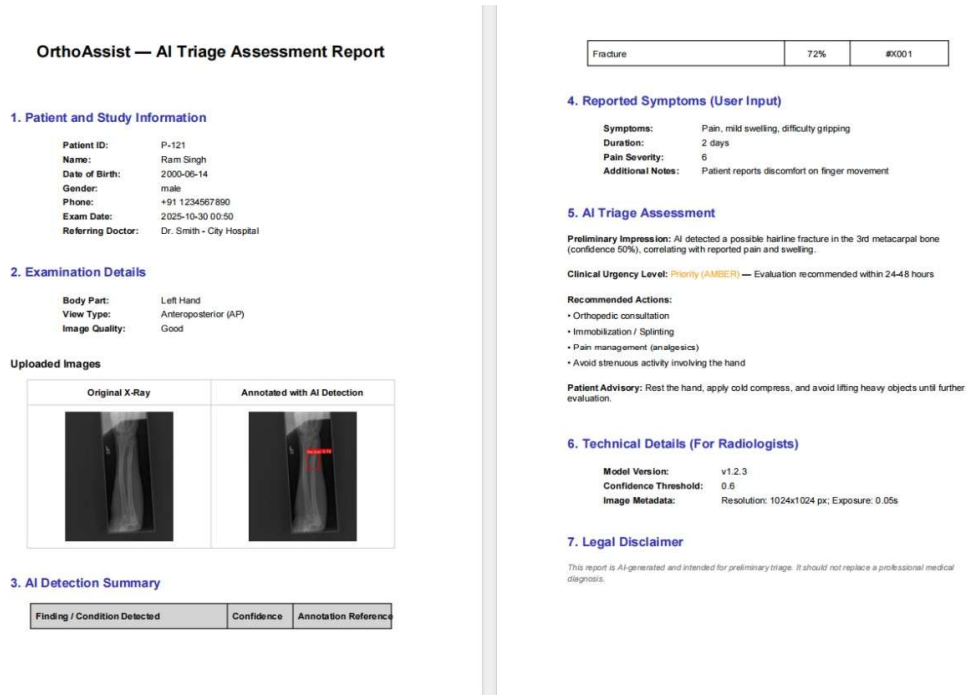


Fig. 6.5 Chat real-time interaction through multimodal input



OrthoAssist — AI Triage Assessment Report

1. Patient and Study Information

Patient ID:	P-121
Name:	Ram Singh
Date of Birth:	2000-06-14
Gender:	male
Phone:	+91 1234567890
Exam Date:	2025-10-30 00:50
Referring Doctor:	Dr. Smith - City Hospital

2. Examination Details

Body Part:	Left Hand
View Type:	Anteroposterior (AP)
Image Quality:	Good

Uploaded Images

Original X-Ray	Annotated with AI Detection

3. AI Detection Summary

Finding / Condition Detected	Confidence	Annotation Reference
Fracture	72%	#X001

4. Reported Symptoms (User Input)

Symptoms:	Pain, mild swelling, difficulty gripping
Duration:	2 days
Pain Severity:	6
Additional Notes:	Patient reports discomfort on finger movement

5. AI Triage Assessment

Preliminary Impression: AI detected a possible hairline fracture in the 3rd metacarpal bone (confidence 50%), correlating with reported pain and swelling.

Clinical Urgency Level: Priority (AMBER) — Evaluation recommended within 24-48 hours

Recommended Actions:

- Orthopedic consultation
- Immobilization / Splinting
- Pain management (analgesics)
- Avoid strenuous activity involving the hand

Patient Advisory: Rest the hand, apply cold compress, and avoid lifting heavy objects until further evaluation.

6. Technical Details (For Radiologists)

Model Version:	v1.2.3
Confidence Threshold:	0.6
Image Metadata:	Resolution: 1024x1024 px; Exposure: 0.05s

7. Legal Disclaimer

This report is AI-generated and intended for preliminary triage. It should not replace a professional medical diagnosis.

Fig. 6.6 Report Generated

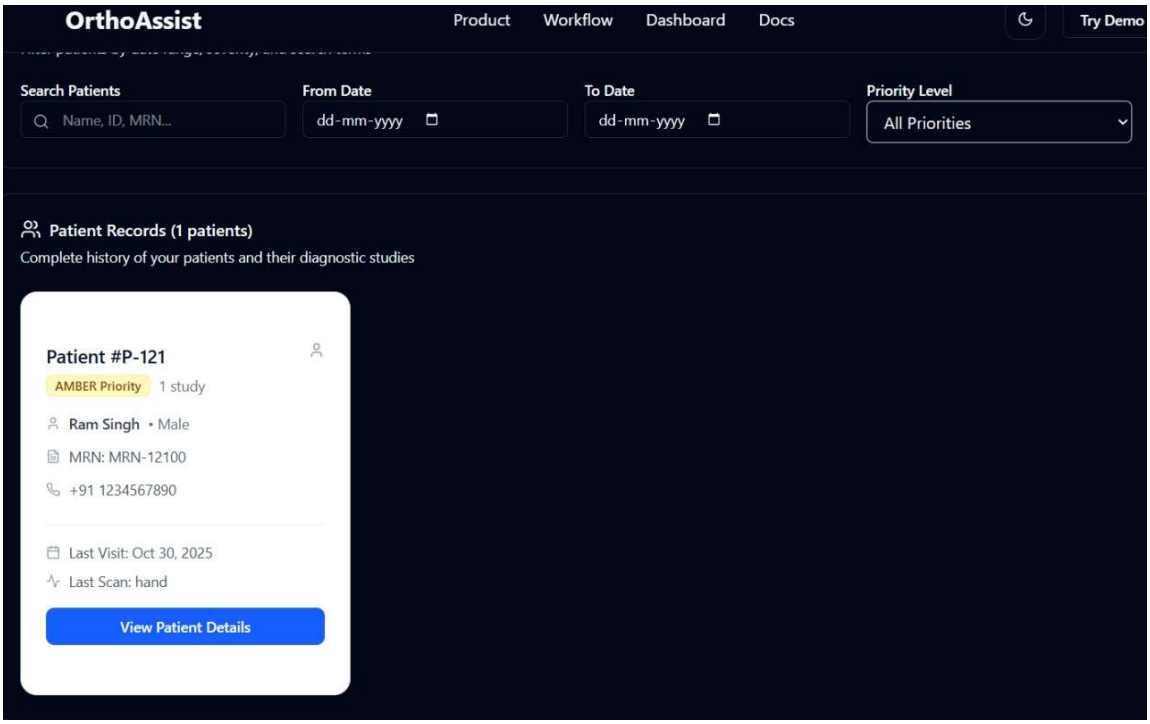


Fig. 6.7 Patient History

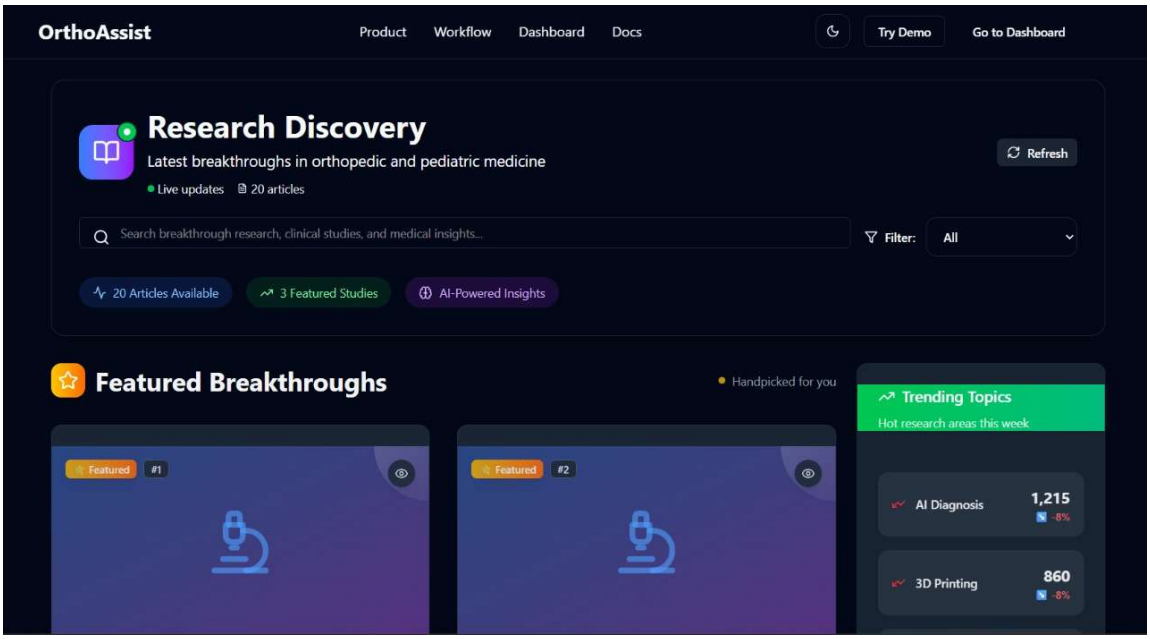


Fig. 6.8 Research Page

Code:

```
class HandAgent:
    """YOLO-based hand fracture detection agent."""

    def __init__(self, model_path: Optional[Path] = None):
        self.device = torch.device("cuda" if torch.cuda.is_available() else "cpu")

        # Hand-specific fracture labels
        self.class_names = [
            "fracture", "displaced_fracture", "hairline_fracture",
            "comminuted_fracture", "avulsion_fracture"
        ]

        self.score_threshold = 0.35
        self.nms_iou = 0.50

    async def detect_fractures(self, image_data: bytes) -> Dict[str, Any]:
        # Run inference with YOLO model
        model_results = self.model.predict(
            image_array,
            conf=self.score_threshold,
            iou=self.nms_iou,
            verbose=False,
            save=False
        )

# Step 1: Body part detection
```

```
detected_body_part = 'hand' if aspect_ratio > 1.2 else 'leg'

# Step 2: Run appropriate YOLO agent
if detected_body_part == 'hand':
    agent = get_hand_agent()
    agent_result = await agent.detect_fractures(image_bytes)
else:
    agent = get_leg_agent()
    agent_result = await agent.detect_fractures(image_bytes)

# Step 3: Generate triage assessment
triage_result = await triage_agent.process_triage_request(
    detections=detection_result['detections'],
    symptoms=request.symptoms,
    body_part=detection_result['body_part']
)

# Step 4: Generate diagnosis summary using LLM
diagnosis_result = await groq_service.generate_diagnosis_summary(
    triage_result=triage_result,
    detections=detection_result['detections'],
    symptoms=request.symptoms
)

async def generate_triage_assessment(
    self, detections: List[Dict], symptoms: Optional[str], body_part: str
) -> Dict[str, Any]:
```

```
    # Build enhanced prompt using medical templates
    user_prompt = MedicalPromptTemplates.build_triage_prompt(
        detections=detections,
        symptoms=symptoms,
        body_part=body_part
    )

    # Make API call with retry logic
    response = await self._make_api_call(
        system_prompt=MedicalPromptTemplates.get_triage_system_prompt(),
        user_prompt=user_prompt,
        model=self.triage_model
    )

def _build_analysis_prompt(self, patient_request: PatientAnalysisRequest) -> str:
    # Sort studies by date for chronological analysis
    sorted_studies = sorted(patient_request.studies,
                            key=lambda x: datetime.fromisoformat(x.date))

    # Analyze temporal progression of findings
    for study in sorted_studies:
        prompt += f"""
        Date: {study_date.strftime('%Y-%m-%d %H:%M')}
        Body Part: {study.bodyPart}
        Triage Level: {study.triage.get('level')}
        Detected Fractures: {len(study.detections)}
        """
```

```
const handleProcess = async () => {
  const formData = new FormData()
  formData.append('file', selectedFile)
  formData.append('processingMode', processingMode)
  formData.append('patientSymptoms', patientSymptoms)

  // Make API call to backend
  const response = await fetch('/api/user/upload', {
    method: 'POST',
    body: formData,
  })

  const result = await response.json()
  setAnalysisResult(result.data)
}
```

Chapter 7: Conclusion

The proposed OrthoAssist – AI-Powered Orthopedic Assistant represents a transformative step in the integration of artificial intelligence, medical imaging, and digital healthcare systems. Designed as a modular and scalable architecture using FastAPI and Next.js, the system combines deep learning, large language models (LLMs), and agent-based coordination to provide a complete orthopedic diagnostic pipeline — from X-ray analysis to report generation.

Through the implementation of YOLO-based object detection, OrthoAssist is capable of identifying fractures in musculoskeletal X-rays with high precision and speed. The use of specialized hand and leg detection agents, coupled with a body part classification mechanism, ensures optimal model routing for accurate anatomical analysis. Additionally, the triage agent dynamically assesses injury severity using clinical rules and AI scoring, providing clear Red/Amber/Green classifications that assist in rapid decision-making for doctors and health professionals.

The LLM-based report generation module, powered by Groq acceleration, converts complex medical data into concise, human-readable summaries and recommendations. This enables clinicians, paramedics, and patients to understand diagnostic outcomes without relying on specialized radiological expertise. The integration with Cloudinary and MongoDB ensures secure, scalable storage of patient data, medical images, and generated reports.

Furthermore, OrthoAssist's agentic MCP framework facilitates asynchronous processing, inter-agent communication, and model context sharing, making the system highly extensible for future medical applications — including orthopedic anomaly segmentation, comparative bone density analysis, and surgical planning assistance. Its real-time feedback, automated analytics dashboard, and intelligent report history tracking make it a powerful tool for both individual practitioners and institutional healthcare systems.

The project demonstrates how AI and multimodal intelligence can significantly enhance orthopedic diagnostics by reducing interpretation delays from hours to minutes. By democratizing expert-level analysis through an accessible digital interface, OrthoAssist bridges the technological divide between urban healthcare centers and rural clinics, empowering medical staff in low-resource settings.

In conclusion, OrthoAssist stands as a proof-of-concept for next-generation orthopedic AI systems that unite efficiency, interpretability, and clinical utility. It has the potential to redefine orthopedic diagnostics by offering fast, explainable, and patient-friendly medical insights while maintaining scalability, modularity, and compliance with healthcare data standards.

Future developments may include expanding the system's dataset diversity, integrating multimodal data (CT, MRI), incorporating federated learning for privacy-preserving AI training, and introducing voice-assisted diagnostic explanations for enhanced accessibility.

Chapter 8: Plagiarism Report

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